Software Requirements Specification

Use Case Based Capstone Template

Document field	Team entry
Project name	Student Savings Project
Team name and members	Ninja Penguins Lupe Campos Joey Contreras Doga Eski Jonathan Ochoa
Sponsor and instructor	Commerce Bank mentors and Syed Jawad Hussain Shah
Version and status	0.3
Date	11/6/2026
Requirements lead	Lupe

Revision history

Version	Date	Author	Change and approval
0.1	9/29/26	Jonathan ochoa	introduction, stakeholders, and product context filled out
0.2	9/30/2026	Doga Eski	Filled out use case model, detailed use cases, functional and nonfunctional requirements, data and interfaces, traceability and change control. Filled remaining placeholders in section 1.4 and 1.5
0.3	11/6/2026	Lupe Campos	Finalized overall content

Contents

1 Introduction

2 Stakeholders Actors and Goals

3 Product Context and Constraints

4 Use Case Model

5 Functional Requirements and Detailed Use Cases

6 Nonfunctional Requirements

7 Data and External Interfaces

8 Traceability and Sprint Mapping

9 Review and Change Control

Appendix A Reusable Detailed Use Case Form

1 Introduction

1.1 Purpose and audience

[State what this document specifies, who should review it, and how it will support design, sprint planning, testing, and sponsor review.]

The purpose of this document is to define the functional and non functional requirements for the project. The document details what the budgeting and loan management application should provide and how users will interact with the application and the quality standards the system should meet.

1.2 Problem and product vision

[Describe the user or business problem without prescribing a solution. Then state one concise product vision.]

Managing personal finances can be difficult when information about spending, loans, accounts, and goals are spread across multiple applications or tracked manually. The product will provide a centralized web application where users can organize their budgeting and loan information, monitor spending, track financial goals, and view their information in an understandable format.

1.3 Scope

In scope	Out of scope
• user registration • user profiles and account settings • monthly budgeting • adding and managing financial accounts • adding and managing loans	• real banking connections • real money movement • production of financial transactions • other capabilities not yet determined

• tracking loan information • personal financial goals • spending summaries • financial notifications • savings plan • help, support, and feedback • AI assisted financial summaries and interaction	

1.4 Definitions and conventions

Term	Definition
Actor	A person, role, device, or external system outside the product boundary that interacts with the product.
Use case	A goal-oriented interaction between an actor and the system.
Scenario	One path through a use case, including the main success path or an alternate or exception path.
Functional requirement	A capability, rule, validation, calculation, or system response that the product must provide.
Nonfunctional requirement	A measurable quality attribute or constraint on how the product operates or is built.
Must	Mandatory requirement.
Should	Desired requirement that may be deferred with an approved reason.
Could	Optional requirement, built only if time allows
Financial Account	Simulated account (checking, savings, credit carrd or cash) that the user creates in the app with a fabricated starting balance. It is not linked to any real bank

1.5 Assumptions dependencies and open questions

ID	Type	Statement	Owner	Target resolution
A-001	Assumption	The application will use simulated or manually entered financial information	Team	sprint 1
A-002	Assumption	PostgreSQL will be used for strong app data	Team	sprint 1 or 2
A-003	Assumption	User passwords will be securely hashed before being stored	Team	sprint 1 or 2

D-001	Dependency	AI technology used by app must be selected and configured. Possible options are Llama, qwen 3, gemma 3, or other	Team	sprint 2
Q-001	Open question	Section 3.1 allows external services for credit score info, but the project charter lists credit score monitoring as out of scope. Should credit score features stay out of scope?	Jonathan Ochoa	Sprint 2 sponsor check-in
Q-002	Open question	Will the selected AI model be self-hosted or used through a hosted API? The answer affects response time (NFR-PERF-003) and privacy (NFR-PRIV-002)	Jonathan Ochoa	sprint 2
Q-003	Open question	Should financial notifications be in-app only or also sent by email	Doga Eski	sprint 3
Q-004	Open question	Does the sponsor agree with the proposed security settings: a 30-minute inactivity timeout and a 15-minute lockout after 5 failed sign-ins?	Joey Contreras	sprint 1
Q-005	Open question	The charter gives two end dates: December 7,2026 and December 11,2026. Which is the final date?	Jonathan Ochoa	sprint 2

2 Stakeholders Actors and Goals

Identify people and external systems before naming use cases. An actor represents a role, not a specific teammate or screen.

2.1 Stakeholders

Stakeholder	Interest or decision	How requirements will be validated
Commerce Bank mentors	Sponsor alignment and technical feedback	Midsemester review, questions, and final walkthrough
Instructor	Course expectations and evaluation	Scheduled reviews and submission feedback

2.2 Actors and user characteristics

Actor ID	Actor or role	Goals	Knowledge permissions and needs
ACT-01	Registered user	Manage budgets, accounts, loans, spending, and financial goals	May have limited financial or technical knowledge. Requires a simple, readable, and easy to navigate interface.

ACT-02	Unregistered user	Create an account and access permitted public info	Can access registration and other public pages.
ACT-03	AI service	provide AI-generated responses or summaries based on information available to application	Must operate within the apps security and privacy requirements
ACT-04	System administrator	Maintain application and assist with appropriate system or support functions	Requires administrative permissions that are separate from normal user permissions

2.3 Actor goal summary

Actor	Goal	Success outcome	Candidate use case
Registered user	Create a monthly budget	Budget is stored for that user and shown in totals	UC-BUD-01
Registered user	Add an account	Account info is stored and displayed	UC-ACC-01
Registered user	Add a loan	Loan info is stored and available for tracking	UC-LOAN-01
Registered user	Track loan payments	Loan payment info is recorded and reflected in loan	UC-LOAN-02
Registered user	Create a financial goal	Financial goal is stored with its target and timeframe	UC-GOAL-01
Registered user	Review spending	Available spending info is summarized for the user	UC-SPEND-01
Registered user	Reset password	User verifies their identity and creates new password	UC-AUTH-01
Unregistered user	Create an account	A new user account is created successfully	UC-AUTH-02
AI service	Generate financial summary	A summary is generated from available app data	UC-AI-01
Registered user	Use financial assistant	User can interact with AI feature and receive a supported response	UC-AI-02

3 Product Context and Constraints

3.1 Product environment and system boundary

[Describe the browser, front end, server or API, database, authentication, external services, and environments at the level needed to understand requirements. Reference the architecture document for design detail.]

The product will be a full stack web app that users access through a web browser. The app will provide functionality for budgeting, loan management, account tracking, financial goals, and other approved financial features.

The system boundary includes the following major components:

- Browser: Users will access the application through a modern web browser on supported desktop or mobile devices.

- Frontend: The application will use a frontend framework or library such as React or Blazor. A CSS framework such as Bootstrap or Tailwind CSS will be used to provide consistent styling and responsive layouts. (subject to change by front end)

- Application Server/API: The server-side application will process user requests, apply business rules, manage authentication, perform calculations, and communicate with the database and approved external services.

- Database: PostgreSQL will be used to store application data, including user profiles, budgets, accounts, loans, goals, and other required application information.

- Authentication: The system will provide user registration and login. Passwords must be securely hashed before being stored and must never be stored as plain text.

- External Services: Approved external services may be used for financial data, credit-score information, or AI functionality when appropriate and available. If an external service cannot be used, the application will support manual entry or other approved alternatives.

- Testing Environment: Development and testing environments will use fabricated or dummy financial information. Real banking credentials, payment-card information, and real financial transactions will not be used.

3.2 Sponsor brief baseline

- Build a full-stack web application selected by the team; do not use a CMS such as WordPress.
- Include a meaningful financial component such as budget tracking, simulated payments, or investments.
- Support user registration and login; store passwords as secure hashes, never as plain text.
- Use a database, a front-end framework or library, and one CSS framework.
- Provide Login/Register, Home/Dashboard, User Profile/Settings, and at least two additional pages.
- Use only fake or dummy card and financial data when simulating transactions.
- Include unit testing; automated tests are strongly encouraged.
- Choose and complete at least two stretch goals from the sponsor brief.

3.3 Chosen stretch goals

Stretch goal	Reason for selection	Use cases or requirements affected	Evidence
AI financial assistant	Provides additional value by allowing users to interact with an AI assistant and receive summaries of available financial information	UC-AI-01, UC-AI-02	Demonstration of the AI chatbox and a test/report showing an AI-generated transaction summary using fabricated data
Personal savings/financial goals	Extends the budgeting functionality by allowing users to establish financial goals and track progress toward a target amount and timeframe.	UC-GOAL-01	Demonstration of creating a financial goal, entering a target amount and timeframe, and displaying goal progress.

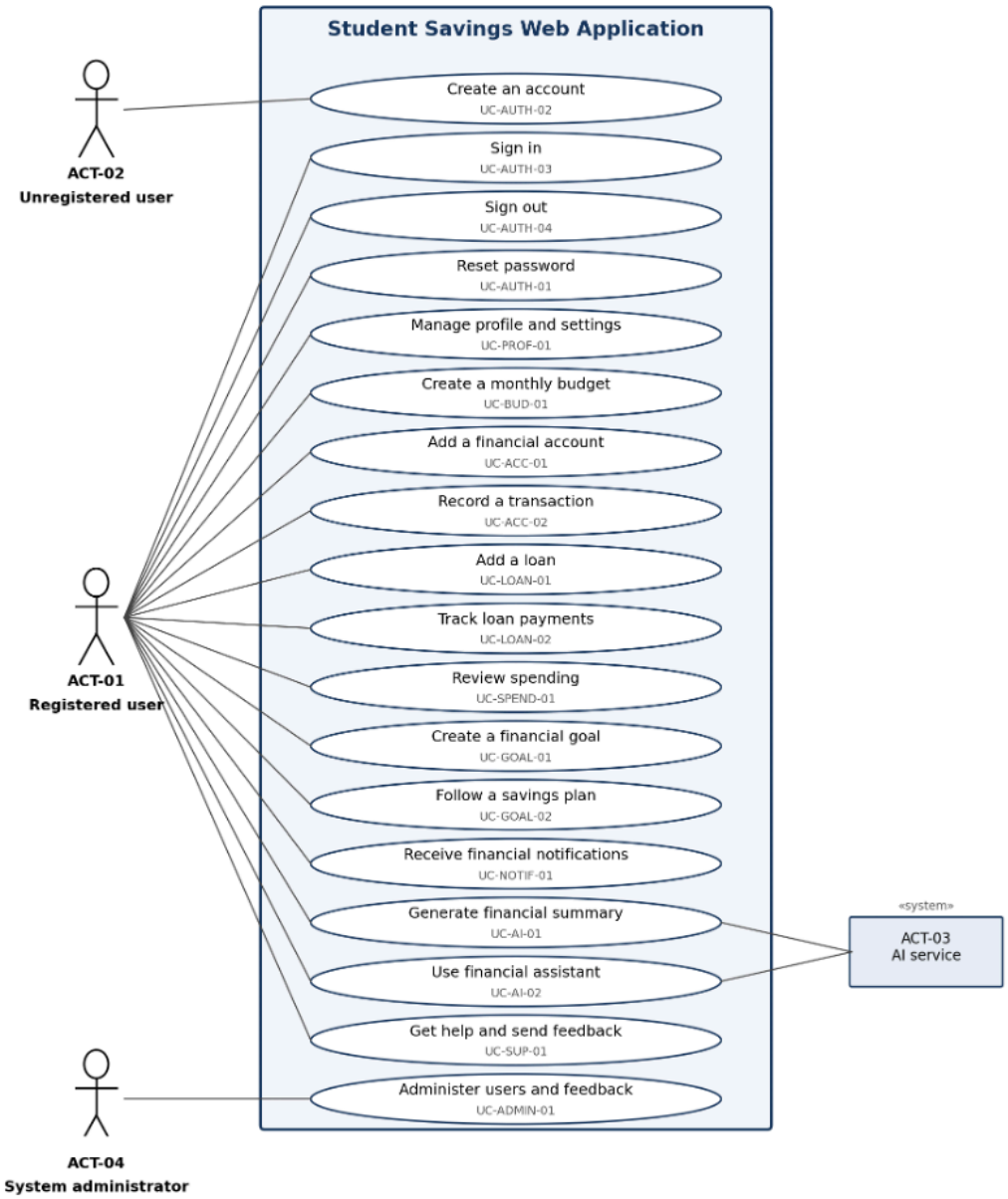
3.4 Business rules

Rule ID	Rule	Source	Related use cases
BR-001	Development, testing, and demonstrations must use fabricated financial and card data.	Sponsor brief	UC-ACC-01
BR-002	User passwords must never be stored as plain text and must be securely hashed before being stored	Sponsor brief	UC-AUTH-01
BR-003	Financial data may be manually entered or imported using an approved method	Project decision	UC-AUTH-01

4 Use Case Model

4.1 Use case diagram

Insert a readable diagram that shows actors outside the system boundary and goal-oriented use cases inside it. Use include or extend only when the relationship improves understanding.



4.2 Use case catalog

ID	Use case name	Primary actor	Priority	Status	Target	Owner
UC-AUTH-01	Reset password	Registered user	Must	Draft	Sprint 1	Joey Contreras
UC-AUTH-02	Create an account	Unregistered user	Must	Draft	Sprint 1	Joey Contreras
UC-AUTH-03	Sign in	Registered user	Must	Draft	Sprint 1	Joey Contreras
UC-AUTH-04	Sign out	Registered user	Must	Draft	Sprint 1	Joey Contreras
UC-PROF-01	Manage profile and settings	Registered user	Must	Draft	Sprint 2	Doga Eski
UC-BUD-01	Create a monthly budget	Registered user	Must	Draft	Sprint 2	Doga Eski
UC-ACC-01	Add a financial account	Registered user	Must	Draft	Sprint 2	Joey Contreras
UC-ACC-02	Record a transaction	Registered user	Must	Draft	Sprint 2	Joey Contreras
UC-LOAN-01	Add a loan	Registered user	Must	Draft	Sprint 3	Joey Contreras
UC-LOAN-02	Track loan payments	Registered user	Must	Draft	Sprint 3	Joey Contreras
UC-UC-SPEND-01	Review spending	Registered user	Must	Draft	Sprint 3	Doga Eski
UC-GOAL-01	Create a financial goal	Registered user	Must (stretch)	Draft	Sprint 3	Lupe Campos
UC-GOAL-02	Follow a savings plan	Registered user	Should	Proposed	Sprint 4	Lupe Campos
UC-NOTIF-01	Receive financial notifications	Registered user	Should	Proposed	Sprint 4	Doga Eski
UC-AI-01	Generate financial summary	AI service (requested by registered user)	Must (stretch)	Draft	Sprint 4	Jonathan Ochoa
UC-AI-02	Use financial assistant	Registered user	Must (stretch)	Draft	Sprint 4	Jonathan Ochoa
UC-SUP-01	Get help and send feedback	Registered user	Should	Proposed	Sprint 4	Lupe Campos
UC-ADMIN-01	Administer users and feedback	System administrator	Should	Proposed	Sprint 4	Jonathan Ochoa

4.3 Use case relationships and scope notes

Relationship or boundary	Explanation and reason
Sign in (UC-AUTH-03) as a precondition	All ACT-01 use cases need an authenticated session. It is listed as a precondition, not an include relationship, so the diagram stays readable
UC-GOAL-02 extends UC-GOAL-01	A savings plan is optional behavior that applies only when a goal has a target date and the user asks for a plan.
UC-LOAN-02 and UC-ACC-02	Recording a loan payment can also record a matching transaction from a chosen account (FR-LOAN-008). This is modeled as a step, not an include relationship.
UC-AI-01 and UC-SPEND-01	The AI summary uses the same totals the spending summary shows. The AI service is the primary actor of UC-AC-01 (section 2.3); the registered user requests the summary.
Out of scope actor goal: link a real bank account	Excluded as a real banking connection (section 1.3) and by BR-001. Manual entry and CSV import of fabricated data (BR-003) are used instead.
Out of scope actor goal: check credit score	Listed as out of scope in the project charter, although section 3.1 allows it. Pending confirmation (Q-001).
Out of scope actor goal: native mobile app	Web only this semester. Phones are supported through responsive design (NFR-USE-002).

5 Functional Requirements and Detailed Use Cases

Use cases are the main source of functional requirements. Sections 5.1 to 5.13 give a detailed form for each Must use case. The Should use cases and Sign out are defined by the brief use cases in section 5.14 and their FR statements in section 5.15. Each will get a detailed form, copied from Appendix A, before its story enters a sprint.

5.1 UC-AUTH-01 Reset password

Field	Team entry
Use case ID and name	UC-AUTH-01 Reset password
Goal	Regain access to the account after forgetting the password.
Primary actor	ACT-01 Registered user
Supporting actors	None (the reset link is delivered by the email service, INT-003)

Trigger	The user selects Forgot password on the sign-in page.
Preconditions	The user is not signed in.
Success postconditions	The password is replaced with a new hash, the reset link no longer works, and all existing sessions are ended. The user can sign in with the new password.
Minimum guarantee	The old password keeps working. The system never reveals whether an email address is registered.
Priority status target owner	Must; Draft; Sprint 1; Joey Contreras

Main success scenario

Step	Actor or system	Action or response
1	Actor	Enters an email address and submits.
2	System	Shows the same confirmation whether or not the email is registered. If it is registered, creates a single-use link that expires in 30 minutes and sends it through the email service.
3	Actor	Actor opens the link from the email and enters a new password twice.
4	System	Checks the link and the password policy (FR-AUTH-003), stores the new hash, disables the link, ends all sessions, and shows Sign in with a success message.

Alternate and exception flows

Flow ID	Related step	Condition	System behavior and recovery
2a	2	The email service is unavailable	Logs the failure without the link. The user sees the normal confirmation and can request another link after 1 minute.
4a	4	The link has expired or was already used	Shows "This link has expired" with an option to request a new one. Returns to step 1.
4b	4	The new password fails FR-AUTH-003, or the two entries do not match	Shows the rule that failed. Returns to step 3.

Rules data quality and evidence

Item	Linked IDs or details
Business rules	BR-002
Data inputs and outputs	DATA-RESET, DATA-USER, DATA-SESSION

Functional requirements	FR-AUTH-003, FR-AUTH-007, FR-AUTH-008, FR-AUTH-009
Nonfunctional requirements	NFR-SEC-001, NFR-SEC-004
Acceptance and test evidence	TC-AUTH-003, TC-AUTH-007 to TC-AUTH-009. Example: Given a reset link created 31 minutes ago, when the user opens it, then the expired message is shown and the password does not change.
Open questions and assumptions	None
Backlog mapping	US-AUTH-06 (main scenario); US-AUTH-07 (flows 2a, 4a, 4b)

5.2 UC-AUTH-02 Create an account

Field	Team entry
Use case ID and name	UC-AUTH-02 Create an account
Goal	Create a personal account so the user's financial information is saved and private.
Primary actor	ACT-02 Unregistered user
Supporting actors	None
Trigger	The unregistered user selects Create account on the Login/Register page.
Preconditions	The user is not signed in.
Success postconditions	An account exists with a hashed password, the User role, and the default categories (FR-BUD-007). The user is signed in and sees an empty Dashboard.
Minimum guarantee	No account or partial record is created. The password never appears in plain text in storage or logs.
Priority status target owner	Must; Draft; Sprint 1; Joey Contreras

Main success scenario

Step	Actor or system	Action or response
1	Actor	Opens the registration form.
2	System	Shows fields for first name, last name, email, password, and confirm password, with a note to enter only fabricated financial data (BR-001).
3	Actor	Fills in the fields and submits.
4	System	Checks required fields, email format, email uniqueness, the password policy (FR-AUTH-003), and that both passwords match.

5	System	Hashes the password, creates the account with default categories, starts a session, and shows the Dashboard with a welcome message.

Alternate and exception flows

Flow ID	Related step	Condition	System behavior and recovery
4a	4	A required field is missing or badly formatted	Shows a message next to each invalid field and keeps what the user entered (except passwords). Returns to step 3.
4b	4	The email is already registered (ignoring case)	Shows "An account with this email may already exist. Sign in or reset your password." Nothing is created. Returns to step 3.
4c	4	The password fails FR-AUTH-003, or the two passwords do not match	Shows the rule that failed. Returns to step 3.
5a	5	The database is unavailable	Shows "We couldn't create your account. Please try again." Saves nothing and logs the error without the password.

Rules data quality and evidence

Item	Linked IDs or details
Business rules	BR-001, BR-002
Data inputs and outputs	DATA-USER, DATA-CAT. Test values: first name "Test", last name "Student", email test.student01@example.com, password "Penguin2026".
Functional requirements	FR-AUTH-001, FR-AUTH-002, FR-AUTH-003, FR-BUD-007
Nonfunctional requirements	NFR-SEC-001, NFR-SEC-002, NFR-SEC-005, NFR-UI-001, NFR-UI-003
Acceptance and test evidence	TC-AUTH-001 to TC-AUTH-003. Example: Given no account exists for test.student01@example.com, when the user registers with valid fields, then an account is created, the stored password is a hash, and the Dashboard is shown.
Open questions and assumptions	A-003
Backlog mapping	US-AUTH-01 (main scenario); US-AUTH-02 (flows 4a to 4c, 5a)

5.3 UC-AUTH-03 Sign in

Field	Team entry

Use case ID and name	UC-AUTH-03 Sign in
Goal	Get secure access to the user's own financial information.
Primary actor	ACT-01 Registered user (ACT-04 System administrator signs in the same way)
Supporting actors	None
Trigger	The user selects Sign in, or opens a protected page while signed out.
Preconditions	An active account exists. The user is not signed in.
Success postconditions	An authenticated session exists, and the Dashboard (or the Admin page for administrators) is shown.
Minimum guarantee	No session is created. Error messages never reveal whether the email or the password was wrong.
Priority status target owner	Must; Draft; Sprint 1; Joey Contreras

Main success scenario

Step	Actor or system	Action or response
1	Actor	Enters email and password and submits.
2	System	Checks the password against the stored hash and checks that the account is active.
3	System	Creates a session that expires after 30 minutes of inactivity, resets the failed-attempt count, and shows the Dashboard.

Alternate and exception flows

Flow ID	Related step	Condition	System behavior and recovery
2a	2	The email or password is wrong	Shows "Email or password is incorrect" and counts the failed attempt. Returns to step 1.
2b	2	5 failed attempts in a row within 15 minutes	Blocks sign-in for that account for 15 minutes and offers the password reset link (UC-AUTH-01).
2c	2	An administrator has disabled the account	Shows "This account is disabled. Contact support." No session is created.
3a	3	The session later times out from inactivity	The next request goes to Sign in with the message "Your session expired."

Rules data quality and evidence

Item	Linked IDs or details
Business rules	BR-001
Data inputs and outputs	DATA-USER (role, status, failed_login_count, locked_until), DATA-SESSION
Functional requirements	FR-AUTH-004, FR-AUTH-005, FR-AUTH-006, FR-AUTH-011
Nonfunctional requirements	NFR-SEC-001 to NFR-SEC-004
Acceptance and test evidence	TC-AUTH-004 to TC-AUTH-006, TC-AUTH-011. Example: Given 5 failed attempts in the past 15 minutes, when the correct password is entered, then sign-in is still refused until the lockout ends.
Open questions and assumptions	Q-004
Backlog mapping	US-AUTH-03 (main scenario, 3a); US-AUTH-04 (flows 2a to 2c)

5.4 UC-PROF-01 Manage profile and settings

Field	Team entry
Use case ID and name	UC-PROF-01 Manage profile and settings
Goal	Keep personal information and settings current, or close the account.
Primary actor	ACT-01 Registered user
Supporting actors	None
Trigger	The user opens the Profile/Settings page.
Preconditions	The user is signed in.
Success postconditions	The changes are saved and shown. If the account was deleted, all of the user's data is removed and the user is signed out.
Minimum guarantee	If saving fails, the previous values are kept.
Priority status target owner	Must; Draft; Sprint 2; Doga Eski

Main success scenario

Step	Actor or system	Action or response
1	Actor	Opens Profile/Settings.
2	System	Shows first name, last name, email, school (optional), and notification preferences (FR-NOTIF-004).
3	Actor	Edits one or more fields and selects Save.

4	System	Validates the fields, saves them, and confirms the changes.

Alternate and exception flows

Flow ID	Related step	Condition	System behavior and recovery
3a	3	The user changes the password	Requires the current password and a new password that meets FR-AUTH-003. Ends the user's other sessions.
3b	3	The user deletes the account	Asks for confirmation and the current password, then permanently deletes the user and all related data and signs the user out.
4a	4	The new email is already used by another account	Shows a message. Nothing is saved. Returns to step 3.
4b	4	A field is invalid	Shows a message next to the field. Returns to step 3.

Rules data quality and evidence

Item	Linked IDs or details
Business rules	BR-002
Data inputs and outputs	DATA-USER (names 1 to 50 characters)
Functional requirements	FR-PROF-001 to FR-PROF-004, FR-AUTH-003, FR-NOTIF-004
Nonfunctional requirements	NFR-PRIV-003, NFR-UI-003, NFR-UI-005
Acceptance and test evidence	TC-PROF-001 to TC-PROF-004
Open questions and assumptions	None
Backlog mapping	US-PROF-01 (main scenario, 4a, 4b); US-PROF-02 (3a); US-PROF-03 (3b)

5.5 UC-BUD-01 Create a monthly budget

Field	Team entry
Use case ID and name	UC-BUD-01 Create a monthly budget
Goal	Plan how much to spend in each category this month.
Primary actor	ACT-01 Registered user

Supporting actors	None
Trigger	The user selects Create budget on the Budget page.
Preconditions	The user is signed in.
Success postconditions	The budget is saved with its category limits. The total is shown, and the spending summary compares spending against it.
Minimum guarantee	No partial budget is saved.
Priority status target owner	Must; Draft; Sprint 2; Doga Eski

Main success scenario

Step	Actor or system	Action or response
1	Actor	Selects a month (the current month by default) and chooses Create budget.
2	System	Lists the user's categories with a limit field for each and offers to copy last month's limits.
3	Actor	Enters limits for one or more categories, optionally adds a custom category, and submits.
4	System	Validates the amounts, saves the budget, and shows the total (the sum of the limits).
5	System	Shows the budget on the Budget page and the Dashboard.

Alternate and exception flows

Flow ID	Related step	Condition	System behavior and recovery
1a	1	A budget already exists for that month	Opens the existing budget for editing instead (FR-BUD-004).
3a	3	An amount is zero or negative, not a number, has more than 2 decimal places, or is over $1,000,000.00	Shows a message next to the field. Nothing is saved. Returns to step 3.
3b	3	No category limit was entered	Shows "Enter at least one category limit." Returns to step 3.
4a	4	The database is unavailable	Shows an error, saves nothing, and keeps the entered values so the user can retry.

Rules data quality and evidence

Item	Linked IDs or details
Business rules	BR-001
Data inputs and outputs	DATA-BUDGET, DATA-CAT. Test values: Groceries $250.00, Transportation $80.00, Dining Out $60.00.
Functional requirements	FR-BUD-001 to FR-BUD-007
Nonfunctional requirements	NFR-ACCU-001, NFR-REL-001, NFR-UI-001, NFR-UI-003
Acceptance and test evidence	TC-BUD-001 to TC-BUD-007. Example: Given no budget for October 2026, when the user saves the three test limits, then the total shows $390.00.
Open questions and assumptions	None
Backlog mapping	US-BUD-01 (main scenario); US-BUD-02 (3a, 3b, 4a); US-BUD-03 (1a, editing, copy last month)

5.6 UC-ACC-01 Add a financial account

Field	Team entry
Use case ID and name	UC-ACC-01 Add a financial account
Goal	Represent where the user's money is kept so balances and spending can be tracked in one place.
Primary actor	ACT-01 Registered user
Supporting actors	None
Trigger	The user selects Add account on the Accounts page.
Preconditions	The user is signed in.
Success postconditions	The simulated account is saved and listed with its current balance.
Minimum guarantee	No partial account is saved. No real account or card number is collected.
Priority status target owner	Must; Draft; Sprint 2; Joey Contreras

Main success scenario

Step	Actor or system	Action or response
1	Actor	Selects Add account.
2	System	Shows fields for account name, type (Checking, Savings, Credit card, or Cash), and starting balance; for Credit card, also a credit limit. A note says to enter fabricated data only.

3	Actor	Fills in the fields and saves.
4	System	Validates the fields, saves the account, and shows the Accounts list with each balance and the total across accounts.

Alternate and exception flows

Flow ID	Related step	Condition	System behavior and recovery
3a	3	The name is empty or longer than 40 characters, or already used by another of the user's accounts	Shows a message next to the field. Returns to step 3.
3b	3	The starting balance is not a valid amount, or a credit limit is zero or negative	Shows a message next to the field. Returns to step 3.
3c	1	The user edits or deletes an account	Saves the change. Deleting asks for confirmation and warns that the account's transactions will also be deleted.

Rules data quality and evidence

Item	Linked IDs or details
Business rules	BR-001, BR-003
Data inputs and outputs	DATA-ACCOUNT. Test values: "Campus Checking", Checking, $1,200.00; "Student Card", Credit card, balance owed $150.00, limit $1,000.00.
Functional requirements	FR-ACC-001 to FR-ACC-004
Nonfunctional requirements	NFR-PRIV-001, NFR-ACCU-001, NFR-UI-005
Acceptance and test evidence	TC-ACC-001 to TC-ACC-004
Open questions and assumptions	A-001
Backlog mapping	US-ACC-01

5.7 UC-ACC-02 Record a transaction

Field	Team entry
Use case ID and name	UC-ACC-02 Record a transaction
Goal	Log income and spending so balances, budgets, and summaries are accurate.

Primary actor	ACT-01 Registered user
Supporting actors	None
Trigger	The user selects Add transaction, or Import transactions, on the Accounts page.
Preconditions	The user is signed in and has at least one financial account.
Success postconditions	The transaction is saved. The account balance, that month's category spending, and the spending summary are updated.
Minimum guarantee	No partial transaction is saved, and balances are unchanged if saving fails.
Priority status target owner	Must; Draft; Sprint 2; Joey Contreras

Main success scenario

Step	Actor or system	Action or response
1	Actor	Selects Add transaction.
2	System	Shows fields for account, type (Expense or Income), amount, date (today by default), category (for expenses), and an optional description.
3	Actor	Fills in the fields and saves.
4	System	Validates the fields, saves the transaction, updates the account balance and category spending, and confirms.

Alternate and exception flows

Flow ID	Related step	Condition	System behavior and recovery
3a	3	The amount is zero or negative or has more than 2 decimal places	Shows a message next to the field. Returns to step 3.
3b	3	The date is in the future	Shows "Transaction date can't be in the future." Returns to step 3.
4a	4	The expense puts its category over its budget limit	Saves the transaction and shows a warning with the amount over the limit.
1a	1	The user imports a CSV file of fabricated transactions (INT-005)	Validates every row, imports the valid rows, and lists rejected rows with their row numbers and reasons.
1b	1	The user edits or deletes a transaction	Saves the change (deleting asks for confirmation) and recalculates balances and totals.

Rules data quality and evidence

Item	Linked IDs or details
Business rules	BR-001, BR-003
Data inputs and outputs	DATA-TXN. Test value: Expense, $42.18, 10/03/2026, Groceries, "Student Card".
Functional requirements	FR-ACC-005 to FR-ACC-010
Nonfunctional requirements	NFR-ACCU-001, NFR-REL-001, NFR-PERF-002
Acceptance and test evidence	TC-ACC-005 to TC-ACC-010. Example: Given Groceries has a $250.00 limit and $230.00 spent, when a $42.18 expense is saved, then the warning shows $22.18 over.
Open questions and assumptions	None
Backlog mapping	US-ACC-02 (main scenario, 3a, 3b, 4a); US-ACC-03 (1b and transaction list); US-ACC-04 (1a CSV import)

5.8 UC-LOAN-01 Add a loan

Field	Team entry
Use case ID and name	UC-LOAN-01 Add a loan
Goal	Record a loan so the user can track it.
Primary actor	ACT-01 Registered user
Supporting actors	None
Trigger	The user selects Add loan on the Loans page.
Preconditions	The user is signed in.
Success postconditions	The loan is saved and listed with its estimated payoff date and the total balance of all loans.
Minimum guarantee	No partial loan is saved.
Priority status target owner	Must; Draft; Sprint 3; Joey Contreras

Main success scenario

Step	Actor or system	Action or response
1	Actor	Selects Add loan.

2	System	Shows fields for loan name, type (federal subsidized, federal unsubsidized, private, or other), original principal, current balance, APR, minimum monthly payment, and next due date.
3	Actor	Fills in the fields and saves.
4	System	Validates the fields, saves the loan, and shows it with its estimated payoff date and total remaining interest (FR-LOAN-004).

Alternate and exception flows

Flow ID	Related step	Condition	System behavior and recovery
3a	3	The APR is outside 0 to 30%, the balance is more than the original principal, or an amount is zero or negative	Shows a message next to the field. Returns to step 3.
4a	4	The minimum payment is less than or equal to one month's interest	Saves the loan and warns that it will never be paid off at this payment.
3b	1	The user edits or deletes a loan	Saves the change (deleting asks for confirmation) and recalculates the payoff estimate.

Rules data quality and evidence

Item	Linked IDs or details
Business rules	BR-001
Data inputs and outputs	DATA-LOAN. Test value: "Fed Unsub Loan A", original $5,500.00, balance $5,500.00, APR 6.53%, minimum $62.00.
Functional requirements	FR-LOAN-001 to FR-LOAN-003
Nonfunctional requirements	NFR-ACCU-001, NFR-MNT-001
Acceptance and test evidence	TC-LOAN-001 to TC-LOAN-003
Open questions and assumptions	Loan calculation method (FR-LOAN-003, FR-LOAN-004) to be confirmed with the sponsor
Backlog mapping	US-LOAN-01

5.9 UC-LOAN-02 Track loan payments

Field	Team entry

Use case ID and name	UC-LOAN-02 Track loan payments
Goal	Keep a loan balance accurate and see progress toward paying it off.
Primary actor	ACT-01 Registered user
Supporting actors	None
Trigger	The user selects Record payment on a loan.
Preconditions	The user is signed in and has at least one active loan.
Success postconditions	The payment is recorded. The balance, payment history, percent paid off, and payoff estimate are updated.
Minimum guarantee	The balance is not changed unless the whole payment is recorded.
Priority status target owner	Must; Draft; Sprint 3; Joey Contreras

Main success scenario

Step	Actor or system	Action or response
1	Actor	Selects a loan and chooses Record payment.
2	System	Shows fields for amount, date, and the financial account the payment came from (optional).
3	Actor	Enters the payment and submits.
4	System	Applies the payment to accrued interest first, then principal (FR-LOAN-004). Updates the balance and payoff estimate and adds the payment to the history. If an account was chosen, records a matching Loan Payments expense on it.
5	Actor	Views the loan details: balance, payment history, percent of principal paid, estimated payoff date, and remaining interest.

Alternate and exception flows

Flow ID	Related step	Condition	System behavior and recovery
3a	3	The amount is zero or negative, or more than the balance plus accrued interest	Shows a message. Returns to step 3.
4a	4	The payment brings the balance to $0.00	Marks the loan Paid off.
4b	4	The database fails partway through	Rolls back the payment, balance, and transaction together and shows an error (NFR-REL-001).

Rules data quality and evidence

Item	Linked IDs or details
Business rules	BR-001
Data inputs and outputs	DATA-PAY, DATA-LOAN, DATA-TXN
Functional requirements	FR-LOAN-004 to FR-LOAN-008
Nonfunctional requirements	NFR-ACCU-001, NFR-REL-001
Acceptance and test evidence	TC-LOAN-004 to TC-LOAN-008. Example: Given a balance of $5,500.00 at 6.53% APR, when a $62.00 payment is recorded, then $29.93 goes to interest, $32.07 goes to principal, and the new balance is $5,467.93.
Open questions and assumptions	None
Backlog mapping	US-LOAN-02 (steps 1 to 4, 3a, 4b); US-LOAN-03 (step 5, 4a)

5.10 UC-SPEND-01 Review spending

Field	Team entry
Use case ID and name	UC-SPEND-01 Review spending
Goal	See at a glance where money is going and how spending compares with the budget.
Primary actor	ACT-01 Registered user
Supporting actors	None
Trigger	The user signs in or opens the Dashboard.
Preconditions	The user is signed in.
Success postconditions	The Dashboard shows the selected month's spending summary. No data is changed.
Minimum guarantee	No data is changed. If a chart fails, the same figures are available as a table.
Priority status target owner	Must; Draft; Sprint 3; Doga Eski

Main success scenario

Step	Actor or system	Action or response
1	Actor	Opens the Dashboard.

2	System	Shows the month's income, spending, and budget remaining; a chart of budget vs. spending by category with over-budget categories highlighted; the change from last month; account balances; total loan balance and next payment due; and goal progress.
3	Actor	Picks a different month or selects a category.
4	System	Updates the view. Selecting a category lists its transactions for the month.

Alternate and exception flows

Flow ID	Related step	Condition	System behavior and recovery
2a	2	There is no budget for the month	Shows spending only and prompts the user to create a budget.
2b	2	The user has no data yet	Shows an empty state with links to Add account, Create budget, and Add transaction.
2c	2	The chart cannot be shown, or the user uses a screen reader	Provides an equivalent data table (NFR-ACC-002).

Rules data quality and evidence

Item	Linked IDs or details
Business rules	BR-001
Data inputs and outputs	Reads DATA-BUDGET, DATA-TXN, DATA-ACCOUNT, DATA-LOAN, DATA-GOAL
Functional requirements	FR-SPEND-001 to FR-SPEND-005
Nonfunctional requirements	NFR-PERF-001, NFR-ACC-001, NFR-ACC-002, NFR-USE-002
Acceptance and test evidence	TC-SPEND-001 to TC-SPEND-005
Open questions and assumptions	None
Backlog mapping	US-SPEND-01 (main scenario, 2c); US-SPEND-02 (widgets, 2a, 2b)

5.11 UC-GOAL-01 Create a financial goal

Field	Team entry
Use case ID and name	UC-GOAL-01 Create a financial goal
Goal	Set a savings target and see progress toward it.

Primary actor	ACT-01 Registered user
Supporting actors	None
Trigger	The user selects New goal on the Goals page.
Preconditions	The user is signed in.
Success postconditions	The goal is stored with its target amount and timeframe, and its progress is shown on the Goals page and the Dashboard.
Minimum guarantee	No partial goal is saved.
Priority status target owner	Must (stretch goal); Draft; Sprint 3; Lupe Campos

Main success scenario

Step	Actor or system	Action or response
1	Actor	Selects New goal.
2	System	Shows fields for goal name, target amount, target date, and amount already saved (optional, $0.00 by default).
3	Actor	Fills in the fields and saves.
4	System	Validates the fields, saves the goal, and shows its progress bar, percent complete, and amount remaining.
5	Actor	Later adds a contribution to the goal.
6	System	Records the contribution and updates the progress. When the saved amount reaches the target, marks the goal Achieved.

Alternate and exception flows

Flow ID	Related step	Condition	System behavior and recovery
3a	3	The target amount is zero or negative, or the target date is not in the future	Shows a message next to the field. Returns to step 3.
3b	3	The amount already saved is more than the target	Shows a message. Returns to step 3.
5a	5	The user edits or deletes the goal	Saves the change (deleting asks for confirmation) and recalculates progress.

Rules data quality and evidence

Item	Linked IDs or details
Business rules	BR-001
Data inputs and outputs	DATA-GOAL. Test value: "Spring break trip", target $600.00, target date 03/01/2027, saved $150.00.
Functional requirements	FR-GOAL-001, FR-GOAL-002, FR-GOAL-003, FR-GOAL-005
Nonfunctional requirements	NFR-ACCU-001, NFR-UI-005
Acceptance and test evidence	TC-GOAL-001 to TC-GOAL-003, TC-GOAL-005. Example: Given the test goal, when it is saved, then it shows 25% complete and $450.00 remaining.
Open questions and assumptions	None
Backlog mapping	US-GOAL-01 (steps 1 to 4, 3a, 3b, 5a); US-GOAL-02 (steps 5, 6)

5.12 UC-AI-01 Generate financial summary

Field	Team entry
Use case ID and name	UC-AI-01 Generate financial summary
Goal	Generate a plain-language summary of the user's month from available app data.
Primary actor	ACT-03 AI service
Supporting actors	ACT-01 Registered user (requests the summary)
Trigger	The registered user selects Summarize my month on the Dashboard or the AI Assistant page.
Preconditions	The user is signed in and has at least one transaction in the selected month.
Success postconditions	A plain-language summary of the month is shown with the disclaimer (FR-AI-003).
Minimum guarantee	No identity data is sent to the AI service, and the user's financial data is not changed.
Priority status target owner	Must (stretch goal); Draft; Sprint 4; Jonathan Ochoa

Main success scenario

Step	Actor or system	Action or response
1	Registered user	Selects Summarize my month.

2	System	Builds an anonymized summary of the month from the user's data: income, spending by category, budget limits, account balances, loan balances, and goal progress. No name or email is included.
3	System	Sends the anonymized data to the AI service.
4	AI service	Returns a plain-language summary that explains the month and points out categories that are over budget.
5	System	Shows the AI-generated summary with the disclaimer and the month it covers.

Alternate and exception flows

Flow ID	Related step	Condition	System behavior and recovery
1a	1	The month has no transactions	Shows "Add some transactions to get a summary" and a link to Add transaction.
3a	3	The AI service is unavailable or takes longer than 15 seconds	Shows a rule-based summary built from the same totals and labels it as an automatic summary (FR-AI-005).
3b	3	The user has reached the daily limit of 20 AI requests	Shows the rule-based summary and the daily limit message.

Rules data quality and evidence

Item	Linked IDs or details
Business rules	BR-001
Data inputs and outputs	Reads DATA-TXN, DATA-BUDGET, DATA-ACCOUNT, DATA-LOAN, DATA-GOAL; writes DATA-AI usage count
Functional requirements	FR-AI-001, FR-AI-003, FR-AI-004, FR-AI-005, FR-AI-006
Nonfunctional requirements	NFR-PRIV-002, NFR-REL-002, NFR-PERF-003
Acceptance and test evidence	TC-AI-001, TC-AI-003 to TC-AI-006. Example: Given a fabricated month where Dining Out is $40.00 over budget, when the user requests a summary, then the summary mentions Dining Out being over budget and shows the disclaimer.
Open questions and assumptions	D-001, Q-002
Backlog mapping	US-AI-01 (main scenario, 1a); US-AI-03 (3a, 3b)

5.13 UC-AI-02 Use financial assistant

Field	Team entry
Use case ID and name	UC-AI-02 Use financial assistant
Goal	Ask a budgeting or loan question and get a plain-language answer.
Primary actor	ACT-01 Registered user
Supporting actors	ACT-03 AI service
Trigger	The user opens the AI Assistant page and types a question or picks a suggested one.
Preconditions	The user is signed in.
Success postconditions	An answer is shown with the disclaimer. The conversation stays visible for the current session only.
Minimum guarantee	No identity data is sent to the AI service, and the user's financial data is not changed.
Priority status target owner	Must (stretch goal); Draft; Sprint 4; Jonathan Ochoa

Main success scenario

Step	Actor or system	Action or response
1	Actor	Opens the AI Assistant page.
2	System	Shows the chatbox, the disclaimer (FR-AI-003), and suggested questions such as "What is the difference between subsidized and unsubsidized loans?"
3	Actor	Enters a question of up to 500 characters and sends it.
4	System	Sends the question and the conversation so far to the AI service, with instructions that keep answers on personal finance.
5	System	Shows the answer with the disclaimer.

Alternate and exception flows

Flow ID	Related step	Condition	System behavior and recovery
3a	3	The question is empty or longer than 500 characters	Shows a message. Returns to step 3.
3b	3	The user has reached the daily limit of 20 AI requests	Shows the limit message and links to the Help page.

4a	4	The AI service is unavailable or takes longer than 15 seconds	Shows a fallback answer matched from the built-in help topics by keyword, or "The assistant is unavailable right now" with a link to Help (FR-AI-005).
5a	5	The question is not about personal finance	The answer politely steers the user back to budgeting, accounts, loans, and goals.

Rules data quality and evidence

Item	Linked IDs or details
Business rules	BR-001
Data inputs and outputs	DATA-AI (conversation kept for the session only)
Functional requirements	FR-AI-002 to FR-AI-007
Nonfunctional requirements	NFR-PRIV-002, NFR-REL-002, NFR-PERF-003, NFR-MNT-003
Acceptance and test evidence	TC-AI-002 to TC-AI-007. Example: Given the AI service is stopped, when the user asks "How do I lower my grocery spending?", then a fallback answer is shown within 17 seconds.
Open questions and assumptions	D-001, Q-002
Backlog mapping	US-AI-02 (main scenario, 3a, 5a); US-AI-03 (4a, 3b)

5.14 Brief use cases

Use case	Summary	Requirements
UC-AUTH-04 Sign out	A signed-in user selects Sign out. The system ends the session and shows the Login page. Using the browser's back button afterward does not show protected data.	FR-AUTH-010; TC-AUTH-010
UC-GOAL-02 Follow a savings plan	For a goal with a target date, the user selects Show savings plan. The system calculates the monthly amount needed (amount remaining divided by months left) and shows whether the user is On track or Behind based on contributions in the last 30 days.	FR-GOAL-004; TC-GOAL-004
UC-NOTIF-01 Receive financial notifications	The system creates in-app notifications when a category reaches 80% and 100% of its budget, when a loan payment is due within 3 days, and when a goal reaches 50% and 100%. The user sees a count badge, can mark notifications read, and can turn each type on or off in Settings.	FR-NOTIF-001 to FR-NOTIF-004; Q-003
UC-SUP-01 Get help and send feedback	The user opens Help and Feedback, reads the FAQ for each main feature, or submits a request (Bug, Question, or Suggestion; 10 to 1,000 characters). The system confirms	FR-SUP-001 to FR-SUP-003

	with a reference number and shows the status of the user's requests.	
UC-ADMIN-01 Administer users and feedback	A system administrator signs in and opens the Admin page. The administrator can view the user list (name, email, join date, status), disable or re-enable an account, and view, respond to, and resolve feedback. The administrator cannot see users' financial records (FR-ADMIN-001). Every admin action is logged.	FR-ADMIN-001 to FR-ADMIN-004; FR-AUTH-011; NFR-SEC-006

5.15 Functional requirements catalog

Each functional requirement links to a use case, business rule, or sponsor constraint. It is verified by the test case with the same number (for example, FR-BUD-003 is verified by TC-BUD-003).

FR ID	Functional requirement	Source or related use cases	Priority	Verification
FR-AUTH-001	The system must allow an unregistered user to create an account with first name, last name, email, and password.	UC-AUTH-02	Must	TC-AUTH-001
FR-AUTH-002	The system must reject registration with an email that is already registered (ignoring case) without creating any record.	UC-AUTH-02 flow 4b	Must	TC-AUTH-002
FR-AUTH-003	The system must require passwords of at least 8 characters with at least one letter and one number at registration, password reset, and password change.	UC-AUTH-01, UC-AUTH-02, UC-PROF-01; BR-002	Must	TC-AUTH-003
FR-AUTH-004	The system must authenticate a user by email and password and start a session.	UC-AUTH-03	Must	TC-AUTH-004
FR-AUTH-005	The system must show a generic message on a failed sign-in and block sign-in for 15 minutes after 5 consecutive failures within 15 minutes.	UC-AUTH-03 flows 2a, 2b	Must	TC-AUTH-005
FR-AUTH-006	The system must end a session after 30 minutes of inactivity and require sign-in for every page except public pages (Login/Register, password reset, and public help).	UC-AUTH-03	Must	TC-AUTH-006
FR-AUTH-007	The system must show the same confirmation for every password reset request, whether or not the email is registered.	UC-AUTH-01	Must	TC-AUTH-007

FR-AUTH-008	The system must email a single-use reset link that expires 30 minutes after it is created.	UC-AUTH-01	Must	TC-AUTH-008
FR-AUTH-009	On a successful reset, the system must store the new password hash, disable the link, and end all of the user's sessions.	UC-AUTH-01	Must	TC-AUTH-009
FR-AUTH-010	The system must end the session on sign-out so that protected pages cannot be viewed afterward.	UC-AUTH-04	Must	TC-AUTH-010
FR-AUTH-011	The system must allow Admin pages and actions only for accounts with the Admin role, and must refuse sign-in to disabled accounts.	UC-AUTH-03 flow 2c, UC-ADMIN-01	Must	TC-AUTH-011
FR-PROF-001	The system must let a user view and edit their first name, last name, email, and school.	UC-PROF-01	Must	TC-PROF-001
FR-PROF-002	The system must require the current password before changing the password, and must end the user's other sessions after the change.	UC-PROF-01 flow 3a	Must	TC-PROF-002
FR-PROF-003	The system must reject an email change to an address already used by another account.	UC-PROF-01 flow 4a	Must	TC-PROF-003
FR-PROF-004	After password confirmation, the system must permanently delete the user's account and all of their data, then sign them out.	UC-PROF-01 flow 3b	Must	TC-PROF-004
FR-BUD-001	The system must let a user create a budget for a calendar month with limits for default and custom categories.	UC-BUD-01	Must	TC-BUD-001
FR-BUD-002	The system must calculate and show a budget's total as the sum of its category limits.	UC-BUD-01	Must	TC-BUD-002
FR-BUD-003	The system must reject a budget amount that is zero, negative, non-numeric, has more than 2 decimal places, or exceeds $1,000,000.00, and must not create a partial record.	UC-BUD-01 flow 3a	Must	TC-BUD-003
FR-BUD-004	The system must allow at most one budget per user per month and open the existing budget when the user tries to create a second one.	UC-BUD-01 flow 1a	Must	TC-BUD-004
FR-BUD-005	The system must let a user edit budget limits and add, rename, or remove custom categories. A category that has transactions cannot be removed until they are moved to another category.	UC-BUD-01	Must	TC-BUD-005

FR-BUD-006	The system should let a user start a new budget by copying the previous month's limits.	UC-BUD-01 step 2	Should	TC-BUD-006
FR-BUD-007	The system must give every new user these default categories: Housing, Groceries, Dining Out, Transportation, Tuition and Books, Loan Payments, Entertainment, and Other.	UC-AUTH-02, UC-BUD-01	Must	TC-BUD-007
FR-ACC-001	The system must let a user add a simulated financial account with a name, type (Checking, Savings, Credit card, or Cash), and starting balance, plus a credit limit for credit cards.	UC-ACC-01	Must	TC-ACC-001
FR-ACC-002	The system must let a user edit or delete an account. Deleting requires confirmation and also deletes the account's transactions.	UC-ACC-01 flow 3c	Must	TC-ACC-002
FR-ACC-003	The system must show each account's current balance (starting balance plus income minus expenses; amount owed for credit cards) and the total across accounts.	UC-ACC-01, UC-SPEND-01	Must	TC-ACC-003
FR-ACC-004	The system must not provide any field for real account numbers, card numbers, or banking credentials.	UC-ACC-01; BR-001	Must	TC-ACC-004
FR-ACC-005	The system must let a user record an income or expense transaction with account, amount, date, category (expenses), and an optional description of up to 200 characters.	UC-ACC-02	Must	TC-ACC-005
FR-ACC-006	The system must reject a transaction with an amount of zero or less, more than 2 decimal places, or a future date.	UC-ACC-02 flows 3a, 3b	Must	TC-ACC-006
FR-ACC-007	The system must let a user edit or delete a transaction and must recalculate balances and totals.	UC-ACC-02 flow 1b	Must	TC-ACC-007
FR-ACC-008	The system must show a warning with the amount over the limit when an expense pushes a category over its budget.	UC-ACC-02 flow 4a	Must	TC-ACC-008
FR-ACC-009	The system should let a user import fabricated transactions from a CSV file in the INT-005 format, importing valid rows and listing rejected rows with reasons.	UC-ACC-02 flow 1a; BR-003	Should	TC-ACC-009
FR-ACC-010	The system must list a user's transactions and let them filter by month, account, and category.	UC-ACC-02, UC-SPEND-01	Must	TC-ACC-010

FR-LOAN-001	The system must let a user add, edit, and delete a loan with name, type, original principal, current balance, APR, minimum monthly payment, and next due date.	UC-LOAN-01	Must	TC-LOAN-001
FR-LOAN-002	The system must reject an APR outside 0 to 30%, a balance above the original principal, and amounts of zero or less.	UC-LOAN-01 flow 3a	Must	TC-LOAN-002
FR-LOAN-003	The system must calculate each loan's estimated payoff date and total remaining interest (interest accrues monthly on the remaining balance at APR / 12; payoff uses the standard amortization formula), and warn when the minimum payment does not cover the monthly interest.	UC-LOAN-01 flow 4a	Must	TC-LOAN-003
FR-LOAN-004	The system must record a loan payment, applying it to accrued interest first and then principal, and update the balance.	UC-LOAN-02	Must	TC-LOAN-004
FR-LOAN-005	The system must reject a payment of zero or less, or one that exceeds the balance plus accrued interest.	UC-LOAN-02 flow 3a	Must	TC-LOAN-005
FR-LOAN-006	The system must show each loan's payment history, percent of principal paid, estimated payoff date, and remaining interest.	UC-LOAN-02 step 5	Must	TC-LOAN-006
FR-LOAN-007	The system must mark a loan Paid off when its balance reaches $0.00.	UC-LOAN-02 flow 4a	Must	TC-LOAN-007
FR-LOAN-008	The system should record a loan payment as a Loan Payments expense on the account the user chooses.	UC-LOAN-02 step 4	Should	TC-LOAN-008
FR-SPEND-001	The Dashboard must show the selected month's total income, total spending, and budget remaining.	UC-SPEND-01	Must	TC-SPEND-001
FR-SPEND-002	The Dashboard must chart budget vs. spending for each category, highlight over-budget categories, and offer an equivalent data table.	UC-SPEND-01	Must	TC-SPEND-002
FR-SPEND-003	The Dashboard must show each category's change in spending from the previous month in dollars and percent.	UC-SPEND-01	Must	TC-SPEND-003
FR-SPEND-004	The Dashboard must show account balances, the total loan balance, the next loan payment due, and goal progress.	UC-SPEND-01	Must	TC-SPEND-004
FR-SPEND-005	The Dashboard must show guidance and links when there is no budget or no data.	UC-SPEND-01 flows 2a, 2b	Must	TC-SPEND-005

FR-GOAL-001	The system must let a user create a financial goal with a name, target amount, target date, and optional amount already saved.	UC-GOAL-01; stretch goal	Must	TC-GOAL-001
FR-GOAL-002	The system must let a user edit and delete a goal and reject a target of zero or less or a target date that is not in the future.	UC-GOAL-01 flows 3a, 5a	Must	TC-GOAL-002
FR-GOAL-003	The system must let a user add contributions to a goal and show the progress bar, percent complete, and amount remaining.	UC-GOAL-01 steps 4 to 6	Must	TC-GOAL-003
FR-GOAL-004	The system should show the monthly amount needed to reach a goal by its target date and whether the user is On track or Behind.	UC-GOAL-02	Should	TC-GOAL-004
FR-GOAL-005	The system must mark a goal Achieved when the saved amount reaches the target.	UC-GOAL-01 step 6	Must	TC-GOAL-005
FR-NOTIF-001	The system should notify a user in the app when a category reaches 80% and 100% of its monthly budget.	UC-NOTIF-01	Should	TC-NOTIF-001
FR-NOTIF-002	The system should notify a user in the app when a loan payment is due within 3 days.	UC-NOTIF-01	Should	TC-NOTIF-002
FR-NOTIF-003	The system could notify a user when a goal reaches 50% and 100% of its target.	UC-NOTIF-01	Could	TC-NOTIF-003
FR-NOTIF-004	The system should show an unread count, let users mark notifications read, and let them turn each notification type on or off in Settings.	UC-NOTIF-01, UC-PROF-01	Should	TC-NOTIF-004
FR-AI-001	The system must generate a plain-language AI summary of a selected month from the user's income, spending by category, budget, balances, loans, and goals.	UC-AI-01; stretch goal	Must	TC-AI-001
FR-AI-002	The system must accept a chat question of 1 to 500 characters, send it with the session's conversation to the AI service, and show the answer.	UC-AI-02; stretch goal	Must	TC-AI-002
FR-AI-003	The system must show the "not financial advice" disclaimer on the AI Assistant page and with every summary and answer.	UC-AI-01, UC-AI-02	Must	TC-AI-003
FR-AI-004	The system must never send a user's name, email, or other identifying data to the AI service. Summaries may include only aggregated amounts and category names.	UC-AI-01, UC-AI-02	Must	TC-AI-004
FR-AI-005	The system must show a rule-based summary or fallback answer when the AI service fails or takes longer than 15 seconds.	UC-AI-01 flow 3a, UC-AI-02 flow 4a	Must	TC-AI-005

FR-AI-006	The system should limit each user to 20 AI requests per day.	UC-AI-01 flow 3b, UC-AI-02 flow 3b	Should	TC-AI-006
FR-AI-007	The system should keep chat history for the current session only and clear it on sign-out.	UC-AI-02	Should	TC-AI-007
FR-SUP-001	The system should provide a Help page with FAQs for each main feature.	UC-SUP-01	Should	TC-SUP-001
FR-SUP-002	The system should let a user submit a feedback or support request (Bug, Question, or Suggestion; 10 to 1,000 characters) and confirm it with a reference number.	UC-SUP-01	Should	TC-SUP-002
FR-SUP-003	The system could show users the status (Open or Resolved) and administrator response for their requests.	UC-SUP-01	Could	TC-SUP-003
FR-ADMIN-001	The system should let an administrator view the user list (name, email, join date, status) without access to users' financial records.	UC-ADMIN-01	Should	TC-ADMIN-001
FR-ADMIN-002	The system should let an administrator disable and re-enable a user account.	UC-ADMIN-01	Should	TC-ADMIN-002
FR-ADMIN-003	The system should let an administrator view feedback, respond to it, and change its status.	UC-ADMIN-01, UC-SUP-01	Should	TC-ADMIN-003
FR-ADMIN-004	The system should log every administrator action with the administrator, action, target, and time.	UC-ADMIN-01	Should	TC-ADMIN-004

6 Nonfunctional Requirements

State measurable quality attributes and constraints that apply across the product or to named use cases. Every NFR must include operating conditions, a target or threshold, and a verification method. Do not write vague statements such as the system must be fast or secure.

ID	Category	Requirement and measure	Verification	Priority
NFR-SEC-001	Security	Passwords must be stored only as salted, one-way hashes; plain-text passwords must not appear in the database, application logs, or error messages.	Database inspection and log review	Must
NFR-SEC-002	Security	After 5 failed sign-in attempts, the account must reject further sign-in attempts for at least 15 minutes.	Security test	Must
NFR-SEC-003	Security	Protected pages and administrative functions must reject unauthenticated or unauthorized requests.	Access-control integration test.	Must

NFR-SEC-004	Security	Password-rest tokens must be single-use, expire after 30 minutes, and become invalid after a reset.	Security and integration test	Must
NFR-SEC-005	Security	Every administrator action must be recorded with administrator identity, action,target, and timestamp, and logs must not be modifiable through the application.	Security and integration test	Must
NFR-PRIV-001	Privacy	The application and demonstrations must use mock financial and card data and must not request real banking credentials or account/card numbers.	Repository database,and UI inspection	Must
NFR-PRIV-001	Privacy	No user's name, email, or other information may be transmitted to the AI service. AI requests may contain only permitted aggregated financial information.	Network/API inspection	Must
NFR-PRIV-002	Privacy	When an account is permanently deleted, all associated profiles, financial, loan, goal, transaction, notification, and chat data must be removed.	Database inspection and deletion test/	Must
NFR-USE-001	Usability	A new user must be able to register, create a monthly budget, add a transaction, and view the Dashboard within 5 minutes without assistance in at least 4/5 usability tests	Usability test	Should
NFR-USE-002	Usability	Validation messages must identify invalid field entries and explain to the user how to correct it.	Manual usability test	Should
NFR-PERF-001	Performance	Dashboard pages must load and display the selected months financial information within 2 seconds for a user up to 10,000 transactions.	Measured performance test	Must
NFR-PERF-002	Performance	Budget, transaction, account loan, and goal calculations must complete within 1 second for datasets containing up to 10,000 transactions.	Measured performance test	Should
NFR-PERF-003	Performance	The AI assistant must display an AI response within 15 seconds when the AI service responds within that time; otherwise, the application must displace required fallback response.	Integration and performance test	Must
NFR-ACC-001	Accessibility	All primary application functions must be operable with only a keyboard, with visible focus indicators on interactive controls.	Manual accessibility test	Must
NFR-ACC-002	Accessibility	Form fields, buttons, charts, and other interactive controls must have accessible labels or equivalent test alternatives.	Accessibility audit	Must
NFR-ACC-003	Accessibility	Dashboard charts must provide an equivalent data table so financial information is available without relying on visual interpretation of the chart.	Manual accessibility test	Must
NFR-REL-001	Reliability	If the AI service fails or exceeds 15 seconds, the application must remain usable and display the rule-based fallback without losing information or financial data from the current session.	Failure-injection test	Must
NFR-REL-002	Reliability	Failed budget, transaction, loan, goal, or account operations must not create partial database records or leave related balances inconsistent.	Transaction rollback and integration test	Must
NFR-REL-003	Reliability	After signing out or session expires, protected pages must remain inaccessible until the user successfully signs in again.	Code review	Should

NFR-MNT-001	Maintainability	Authentication, budgeting, accounts, loans, goals, notifications, AI, and administration functionality must be separated into independently maintainable modules/components.	Code review	Should
NFR-MNT-002	Maintainability	Business rules such as password requirements, session timeout, AI request limits, and budget limits must be configurable without modifying unrelated applications components.	Code/configuration review	Should
NFR-MNT-003	Maintainability	All automated tests must pass before a production can be created with at least 80% statement coverage for core business logic.	CI pipeline and coverage report	Should

6.1 Additional interface quality requirements

NFR ID	Quality requirement and measure	Related use cases	Verification
NFR-UI-001	For the selected representative workflows, all interactive controls must be reachable and operable by keyboard without using a mouse.	UC-AUTH, UC-BUDGET, UC-TRANSACTION, UC-GOAL	Keyboard walkthrough
NFR-UI-002	Labels, validation messages, and navigation must be understandable to users completing the selected task with at least 4/5users completing each selected task without assistance.	UC-AUTH, UC-BUDGET, UC-TRANSACTION, UC-LOAN, UC-GOAL	Usability test with 80% success criterion
NFR-UI-003	Form validation messages must identify the invalid input and explain the required correction within the same form without requiring the user to navigate to another page.	UC-AUTH, UC-BUDGET, UC-TRANSACTION, UC-LOAN, UC-GOAL	Manual interface test
NFR-UI-004	Dashboard charts must provide an equivalent data table containing the same budge and spending information so that the information does not depend on visual chart interpretation.	UC-DASHBOARD	Accessibility inspection and comparison test
NFR-UI-005	Interactive controls must have visible focus indicators and all form fields must have an associated descriptive label.	UC-AUTH, UC-BUDGET, UC-TRANSACTION, UC-LOAN, UC-GOAL, UC-SETTINGS	Accessibility audit
NFR-UI-006	Important confirmation, warning, and error messages must remain visible long enough for a user to read them and must not disappear before 5 seconds unless the user dismisses them.	UC-BUDGET, UC-TRANSACTION, UC-LOAN, UC-GOAL, UC-NOTIFICATION	Manual interface test

7 Data and External Interfaces

7.1 Data requirements

Data ID	Entity or data	Important fields and validation	Ownership privacy and retention	Related use cases
DATA-USER	User account	User ID (unique), first name, last name, email (required, unique, case-insensitive), password(stored only as a salted hash)	Owned by the user; personal and sensitive authentication data. Passwords must never be stored in plain text. All account data must be deleted when the user permanently deletes their account.	UC-REGISTER, UC-LOGIN, UC-PROFILE, UC-RESET-PASSWORD, UC-DELETE-ACCOUNT
DATA-BUDGET	Monthly Budget	Budget ID, user ID, calendar month, category limits, total budget. One budget per user per month. Category amounts must be positive, numeric, have no more than 2 decimal places, and not exceed $1,000,000.00.	Owned by the user; contains private financial planning information. Retained while the account exists and deleted with the account.	UC-CREATE-BUDGET, UC-EDIT-BUDGET, UC-VIEW-DASHBOARD
DATA-CATEGORY	Budget and transaction categories	Category ID, category name, user ID, category type. New users receive the eight default categories. Custom category names must be provided. Categories with associated transactions cannot be deleted until those transactions are reassigned.	User-specific financial classification data. Deleted with the associated account and its data.	UC-MANAGE-CATEGORIES, UC-CREATE-BUDGET, UC-ADD-TRANSACTION
DATA-ACCOUNT	Financial account	Account ID, user ID, account name, account type (Checking, Savings, Credit card, or Cash), starting balance, credit limit (for credit cards), calculated current balance.	Owned by the user; contains simulated financial information only. Real account numbers, card numbers, and banking credentials must not be collected. Deleted with the account and associated transactions.	UC-MANAGE-ACCOUNTS, UC-VIEW-DASHBOARD, UC-ADD-TRANSACTION
DATA-TRANSACTION	Income and expense transaction	Transaction ID, account ID, amount, transaction type, date, category, optional description (maximum 200 characters). Amount must be greater than zero, have no more than 2 decimal places, and date cannot be in the future.	Owned by the user; private financial activity data. Retained while the account exists and deleted when the user or associated account deletes it.	UC-ADD-TRANSACTION, UC-EDIT-TRANSACTION, UC-IMPORT-CSV, UC-VIEW-TRANSACTIONS
DATA-LOAN	Loan	Loan ID, user ID, loan name, loan type, original principal, current balance, APR (0–30%), minimum monthly payment, next due date. Balance cannot exceed original principal, and required amounts must be positive.	Owned by the user; contains private debt and repayment information. Deleted with the user's account.	UC-MANAGE-LOANS, UC-VIEW-LOAN, UC-VIEW-DASHBOARD
DATA-LOANPAY	Loan payment	Payment ID, loan ID, payment amount, payment date, interest paid, principal paid, resulting loan balance. Payment must be greater than zero and cannot exceed the balance plus accrued interest.	Owned by the user; private loan repayment history. Retained with the associated loan and deleted with the user's account.	UC-RECORD-LOAN-PAYMENT, UC-VIEW-PAYMENT-HISTORY

DATA-GOAL	Financial goal	Goal ID, user ID, goal name, target amount, target date, saved amount, progress status. Target amount must be greater than zero, and target date must be in the future when created or updated.	Owned by the user; private financial planning data. Deleted with the user's account.	UC-CREATE-GOAL, UC-EDIT-GOAL, UC-ADD-CONTRIBUTION, UC-VIEW-DASHBOARD
DATA-NOTIF	Notification	Notification ID, user ID, notification type, message, creation date, read status, notification preferences. Notification types include budget thresholds, loan due dates, and goal milestones.	Visible only to the owning user. Retained while relevant to the account and deleted with the user's account.	UC-VIEW-NOTIFICATIONS, UC-MANAGE-NOTIFICATION-SETTINGS
DATA-AI	AI requests and chat history	Request ID, session ID, user question (1–500 characters), conversation context, generated response, selected month, aggregated financial data. Identifying user information must not be sent to the AI service.	Chat history is limited to the current session and cleared on sign-out. AI requests may contain only permitted aggregated financial data and category names.	UC-AI-SUMMARY, UC-ASK-AI
DATA-SUPPORT	Support and feedback request	Request ID, user ID, request type (Bug, Question, or Suggestion), message (10–1,000 characters), submission date, reference number, status (Open or Resolved), administrator response.	Owned by the submitting user and accessible to authorized administrators. Contains potentially personal support information. Deleted with the user's account.	UC-SUBMIT-FEEDBACK, UC-VIEW-SUPPORT-STATUS, UC-ADMIN-MANAGE-FEEDBACK
DATA-ADMINLOG	Administrator audit log	Log ID, administrator ID, action, target, timestamp. Each administrator action must generate a log entry containing all required fields.	Restricted to authorized administrative or audit access. Must not expose users' financial records. Retention period should be defined by the project.	UC-ADMIN-MANAGE-USERS, UC-ADMIN-MANAGE-FEEDBACK, UC-VIEW-AUDIT-LOG

7.2 External interfaces

Interface ID	System or service	Purpose and contract	Authentication data and failure behavior	Owner
INT-001	[API, reporting tool, email service, or browser]	[Inputs, outputs, protocol, version]	[Trust, timeout, retry, unavailable behavior]	[Name]

8 Traceability and Sprint Mapping

8.1 Requirements traceability matrix

Use case or requirement	Scenario or statement	Story task	Design component	Test case	Status
UC-BUD-01	Main flow steps 1-5	US-BUD-04	Budget form API service and repository	TC-BUD-001	[Status]
UC-BUD-2	Alternate flow 3a	US-BUD-05	Validation rules and accessible message	TC-BUD-006	[Status]
UC-AUTH-01	Main flow — User enters first name, last name, email, and valid password to create an account.	US-AUTH-01	Registration form, authentication API service, and user repository	TC-AUTH-001	In progress
UC-AUTH-01	Alternate flow — Registration is rejected when the email is already registered, ignoring case.	US-AUTH-02	Registration validation and user repository	TC-AUTH-002	In progress
UC-AUTH-01	Alternate flow — Registration is rejected when the password is fewer than 8 characters or does not contain at least one letter and one number.	US-AUTH-03	Password validation and registration form	TC-AUTH-003	In progress
UC-AUTH-02	Main flow — User enters a registered email and correct password and is successfully signed in with a new session.	US-AUTH-04	Login form, authentication API service, session service	TC-AUTH-004	In progress
UC-AUTH-02	Alternate flow — User enters an incorrect email or password and receives a generic sign-in failure message.	US-AUTH-05	Login validation and authentication service	TC-AUTH-005	In progress
UC-AUTH-02	Alternate flow — User is blocked from signing in for 15 minutes after 5 consecutive failed attempts within 15 minutes.	US-AUTH-06	Authentication service and login-attempt tracking	TC-AUTH-006	In progress

UC-AUTH-02	Alternate flow — User session ends after 30 minutes of inactivity and sign-in is required to access protected pages.	US-AUTH-07	Session management and authentication middleware	TC-AUTH-007	In progress

8.2 Sprint backlog mapping

Story or task	Use case slice	Acceptance criteria	Target sprint	Owner	Result
US-BUD-01	UC-BUD-01 main success scenario	Given a registered user is signed in, when they create and submit a monthly budget with spending limits for each category, then the budget is saved successfully and the total budget and spending summary are displayed.	Sprint 4	Doga Eski Joey Contreras	Finished
US-AUTH-01	UC-AUTH-01 main success scenario	Given an unregistered user provides a first name, last name, unique email, and valid password, when they submit registration, then a new account is created successfully.	Sprint 1	Doga Eski Joey Contreras	Finished
US-AUTH-02	UC-AUTH-01 alternate flow — duplicate email	Given an email is already registered, when the user attempts to create an account with that email regardless of letter case, then registration is rejected and no new account is created.	Sprint 6	Doga Eski	Planned
US-AUTH-03	UC-AUTH-01 alternate flow — invalid password	Given a password is fewer than 8 characters or does not contain at least one letter and one number, when the user submits registration, then the account is not created and a validation message is displayed.	Sprint 5	Doga Eski	Planned
US-AUTH-04	UC-AUTH-02 main success scenario	Given a registered user provides the correct email and password, when they submit the sign-in form, then the system authenticates the user and starts a session.	Sprint 2	Doga Eski Joey Contreras	Finished
US-AUTH-05	UC-AUTH-02 alternate flow — failed sign-in	Given a user enters an incorrect email or password, when they attempt to sign in, then the system displays a generic failure message without revealing which credential was incorrect.	Sprint 2	Doga Eski	Finished
US-AUTH-06	UC-AUTH-02 alternate flow — sign-in lockout	Given a user has 5 consecutive failed sign-in attempts within 15 minutes, when another sign-in	Sprint 6	Doga Eski	Planned

		attempt is made, then the system blocks sign-in for 15 minutes.			
US-AUTH-07	UC-AUTH-02 alternate flow — session timeout	Given a user has been inactive for 30 minutes, when they attempt to access a protected page, then the system ends the session and requires the user to sign in again.	Sprint 7	Doga Eski	Planned

8.3 Prioritization

Explain how Must, Should, and Could priorities affect the MVP and sprint order. Mandatory sponsor requirements must be completed before optional stretch work. Record architecture or risk-driven sequencing when it differs from business priority.

9 Review and Change Control

9.1 Review checklist

- [ ] Every actor has at least one goal and every in-scope goal maps to a use case.
- [ ] Each detailed use case has a trigger, preconditions, main success scenario, alternate or exception flows, and postconditions.
- [ ] The use case describes behavior and outcomes without unnecessary screen or database design.
- [ ] Functional requirements describe testable behavior and use stable FR identifiers when standalone statements are needed.
- [ ] Nonfunctional security, privacy, usability, performance, accessibility, and reliability requirements include measurable targets and verification methods.
- [ ] Use cases, FRs, and NFRs link to sprint stories and test cases by stable identifier.
- [ ] All examples, fixtures, screenshots, and demonstrations use fabricated financial data.

9.2 Change requests and decisions

Change ID	Requested change and reason	Affected IDs	Decision and approver	Date sprint
CHG-001	[Request and business or technical reason]	[UC BR NFR DATA IDs]	[Approved deferred rejected and by whom]	[Date / sprint]

Appendix A Reusable Detailed Use Case Form

Copy this entire appendix section for each use case. Replace every bracketed prompt and delete rows that do not apply only after the team confirms they are unnecessary.

Field	Team entry
Use case ID and name	[UC-AREA-## Goal-oriented verb phrase]
Goal	[Actor value]
Primary actor	[Actor ID and role]
Supporting actors	[Roles or external systems]
Trigger	[Starting event]
Preconditions	[Required prior state]
Success postconditions	[State after success]
Minimum guarantee	[State preserved after failure]
Priority status target owner	[Entries]

Main success scenario

Step	Actor or system	Action or response
1	[Actor/System]	[Action or observable response]
2	[Actor/System]	[Action or observable response]
3	[Actor/System]	[Action or observable response]
4	[Actor/System]	[Action or observable response]
5	[Actor/System]	[Action or observable response]
6	[Actor/System]	[Action or observable response]

Alternate and exception flows

Flow ID	Related step	Condition	System behavior and recovery

__a	__	[Condition]	[Behavior and return point]
__b	__	[Condition]	[Behavior and return point]
__c	__	[Condition]	[Behavior and return point]

Rules data quality and evidence

Item	Linked IDs or details
Business rules	[BR IDs]
Data inputs and outputs	[Data IDs and validation]
Functional requirements	[FR IDs]
Nonfunctional requirements	[NFR IDs]
Acceptance and test evidence	[TC IDs and acceptance examples]
Open questions and assumptions	[Q and A IDs]
Backlog mapping	[Story or task IDs with scenario slices]

Use case review record

Review field	Entry
Owner and last updated	[Name and date]
Reviewers	[Requirements reviewer, developer, tester, sponsor or instructor when applicable]
Status and target	[Proposed / Draft / Approved; target sprint or release]
Decision or follow-up	[Approved, changes requested, deferred question, and linked issue IDs]

[ ] The primary actor and goal are clear.

[ ] Success, alternate, and failure behavior are testable.

[ ] Business rules, functional requirements, nonfunctional requirements, sprint work, and tests are linked by ID.

[ ] The use case is ready for the target sprint only after review issues are resolved.